# Recognizing epistemic EFX allocations of goods

A short proof for catalogue problem D63-653

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An allocation $A = (A_1, \dots, A_n)$ is an ordered partition of a finite set of goods $M$. Agent $i$ has an additive valuation $v_i(S) = \sum_{g \in S} v_i(g)$, explicitly encoded in binary. Following Caragiannis et al. [1, Definitions 1–2], $A$ is *epistemic EFX* (EEFX) if, for each $i$, there is an allocation $Y^{(i)}$ with $Y_i^{(i)} = A_i$ such that

$$v_i(A_i) \geq v_i\big(Y_j^{(i)} \setminus \{g\}\big) \qquad \text{for every } j \neq i \text{ and } g \in Y_j^{(i)} \text{ with } v_i(g) > 0.$$

Different agents may use different witnesses. The final paragraph of [1, Section 4] asks for the complexity of recognizing such an allocation of goods. Its Section 6 treats chores separately.

**Theorem 1.** *Deciding whether a supplied allocation is EEFX is NP-complete even for exactly three agents with strictly positive integer item values, nonempty supplied bundles, and a singleton supplied bundle for agent 1.*

*Proof.* **Membership in NP.** A certificate supplies one witness allocation for each agent. With $t$ goods it contains $nt$ recipient labels, each using $O(\log n)$ bits. Check completeness, the fixed own bundle, and all EFX inequalities. There are polynomially many checks, and binary integer sums and comparisons take polynomial time. For nonnegative rational values in the unrestricted problem, a common denominator has polynomial bit length, so the same verifier remains polynomial.

**Reduction.** Use the NP-complete problem Partition [2]: given positive integers $a_1, \dots, a_m$ with sum $2B$, ask whether some subset sums to $B$. This restricted source remains NP-complete. Indeed, a signed equal-partition instance is unchanged by taking absolute values and discarding zero entries; doubling all remaining entries makes the total even. An all-zero input can be replaced by $(1, 1)$. Thus $m, B \geq 1$.

Put $H = B + 1$ and $V = mH + B$. Introduce $m$ regular goods $g_1, \dots, g_m$, $m + 2$ dummy goods $z_1, \dots, z_{m+2}$, and a good $d$. Agent 1's item values are

$$v_1(g_\ell) = H + a_\ell, \qquad v_1(z_r) = H, \qquad v_1(d) = V.$$

Supply the allocation

$$A_1 = \{d\}, \qquad A_2 = \{g_1, \dots, g_m, z_2, \dots, z_{m+2}\}, \qquad A_3 = \{z_1\}.$$

For each $j = 2, 3$, let $L = 2m + 3$ and set $v_j(g) = L$ if $g \in A_j$, and $v_j(g) = 1$ otherwise. Agent $j$ values its own bundle at least $L$ and its entire complement at most $L - 1$. Therefore $A$ itself is an EEFX witness for each of these two agents. Every item value is positive.

**Correctness.** For positive additive values, EFX against a nonempty bundle $C$ is equivalent to

$$v_1(C) - \min_{g \in C} v_1(g) \leq V, \tag{1}$$

because deleting a minimum-valued good leaves the largest residual. An agent-1 witness partitions the $2m + 2$ outside goods into two bundles. A bundle with at least $m + 2$ goods leaves at least $m + 1$ goods after any one deletion, giving a residual of at least $(m+1)H > mH + B = V$. Hence both bundles have at most $m + 1$ goods, and counting forces each to have exactly $m + 1$.

There are only $m$ regular goods, so each bundle contains a dummy and has minimum value $H$. If $S_j$ indexes its regular goods, its residual is

$$mH + \sum_{\ell \in S_j} a_\ell.$$

Condition (1) bounds both regular sums by $B$. Together they sum to $2B$, so both equal $B$, giving a partition solution.

Conversely, divide the regular goods according to a partition solution, and pad each side to $m+1$ goods with dummies. Each side needs at least one dummy; the two sides need exactly $m+2$ in total. Their minimum value is $H$, and both residuals equal $mH + B = V$. They give a witness for agent 1, and the auxiliary witnesses already established show that $A$ is EEFX.

**Size.** The construction has $2m + 3$ goods and three valuation rows. Since $a_\ell \leq 2B$, every value uses $O(\log(m+1) + \log(B+1))$ bits. Constructing the explicit allocation and valuation matrix therefore takes polynomial time. This proves the claim. □

## Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

## References


[1] I. Caragiannis, J. Garg, N. Rathi, E. Sharma, and G. Varricchio. *New Fairness Concepts for Allocating Indivisible Items*. arXiv:2206.01710v3, 14 January 2025. Definitions 1–2; Section 4, final paragraph. `https://arxiv.org/html/2206.01710v3`.

[2] R. M. Karp. *Reducibility Among Combinatorial Problems*. In *Complexity of Computer Computations*, Plenum Press, 1972, pp. 85–103. Section 4: Main Theorem (p. 94), problem 20 (p. 97), and the Knapsack-to-Partition reduction (p. 100). doi:10.1007/978-1-4684-2001-2_9. Original-page scan in the 2010 reprint.